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M 304 AUXILIARY STEAM SYSTEM

1 SCOPE OF WORK

The Bidder shall supply a complete integrated auxiliary steam system including but not limited to the following:

- One (1) new Auxiliary Boiler will supply the auxiliary steam to new desalination plant. The auxiliary steam piping complete with valves, thermal insulation and lagging, vents, drains and supports and all other appurtenances.

2 CODES AND STANDARDS

The Bidder shall comply with the requirements of part 4 Subsection 2.5 Codes and Standards.

The Codes and Standards of the following organization, extracted from Part 4 Subsection 2.5 “Codes and Standards” or other equivalent international standards are specifically applicable to the design, manufacture and testing of the equipment included in this specification.

ASME	American Society of Mechanical Engineers
AISC	American Institute of Steel Construction Manual
ANSI	American National Standard Institute, Inc
ASTM	American Society of Testing and Materials
AWS	American Welding Society
EPA	Environmental Protection Agency
NFPA	National Fire Protection Association
DIN	Deutsches Institut Für Normung

3 DESIGN AND OPERATING CONDITIONS

The auxiliary steam system shall be supplied from the combined cycle low pressure steam header to provide steam for the deaerator, sealing, air ejector, and all other equipment requiring steam for operation of block 2, 3 & 4.

For initial start-up purposes of the MSF type Desalination plant block 3 and 4 the supply of auxiliary steam shall be from new Auxiliary Boiler.

the existing Auxiliary Boiler shall be connected to new Desalination Plant to backup the auxiliary steam for initial start-up.

4 DETAILED REQUIREMENTS

The capacity, design pressure and temperature of the auxiliary steam system shall be of the Bidder proven design.

Bidder shall provide automatic isolation and safety valves to protect the auxiliary steam system from overpressure.

The auxiliary steam pressure shall be indicated in the central control room.

Condensate from the auxiliary steam system shall be routed to the condensate collecting tank.

The auxiliary steam piping shall be in accordance with ANSI B31.1, latest edition including addenda.

Piping, support, insulation, valves, and control valves shall be in accordance with Part 4, Specifications M-001, M-002, M-003, M-004, M-005.

4.1 Auxiliary Boiler

The bidder shall use new auxiliary boiler for start-up new desalination system

5 PERFORMANCE TESTING AND GUARANTEES

After equipment is operational, the Bidder shall run a test to ascertain that the system is performing properly in every respect. Any deficiencies brought out by this test shall be promptly corrected by the Bidder at no cost to the Owner.

6 QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality assurance/quality control shall conform to the requirements of Part 4, subsection 2.7.

The quality assurance/quality control documentation shall include but not be limited to the following:

- Material Certifications
- NDE Reports
- All other documentation required by applicable codes and standards.

7 INSTRUMENTATION AND CONTROLS

The Bidder shall be the responsible for the design, supply and installation of complete instrumentation and control systems and devices for the auxiliary steam system and Auxiliary Boiler.

The Bidder shall furnish all the necessary accessories for motor operated valves.

The motor operated valve control switches shall be easily accessible.

The Bidder shall also furnish and install all necessary devices to interface with other control systems such as the Plant Monitoring System and on/off -interlock control system.

8 ELECTRICAL SPECIFICATIONS

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for auxiliary steam system and Auxiliary Boiler.

The electrical equipment and/or accessories shall include but not be limited to, the following

- a. Motors
- b. Local starter enclosures
- c. Conduit, cables and wiring

- d. All electrical control, monitoring and instrumentation devices and other appurtenances.

All electrical equipment shall be suitable for outdoor installation.

All electrical equipment or devices and its installation located in hazardous areas shall comply with IEC 79 and NEPA 70 (NEC) Article 500.

8.2 Motors

For detailed electrical requirements, see Part 4, Technical Specification E-210 Low Voltage Motors.

8.3 Conduits, Cables and Wiring

All conduits, cables, wiring and its installation shall meet the requirements of Part 4. Section 3 E-230 and E-232.

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